Descriptive Logic II: Names, Samples, Simples and Rules

Philosophical Investigations §§33–64, in Phient's terms

Phiano Project · `bin/squares_game`, `bin/language_game mistakes` · 2026-09-30

> In Phient's world, a word (w) is nothing but what it relates to, under what conditions (c). The Phical paradigm, X ⇒ Y, drives and composes Phient's Phasor Language Models: Phinet, Phiano, PhiTon. Phical stores no point: a word is 32–64 relations, each a complex phasor on a closed circle with a count; meaning is a side effect of pairs of words moving under a condition.

This is the second chapter of **Descriptive Logic**. Chapter I (paper 19) read §§1–32 next to Phient's paradigm and played the builders' game. This chapter reads §§33–64 and builds two more games:

- **The squares game** of §48, with the composite names of §64. It tests whether a word's condition can be set by the words before it.
- **The builders' game again, with a teacher who makes mistakes** (§54). It tests whether a rule can be read off practice that contains errors.

As in Chapter I, Wittgenstein is paraphrased and cited by remark number, never quoted. The translation is still under copyright. What this chapter says a remark says is our reading of it.

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1. Five more definitions

Chapter I gave five definitions:

- **D1 use:** a word's meaning is how it moves the answer from what the condition alone says.
- **D2 post:** where it moves the answer enough.
- **D3 game:** which aspects carry from one game to another.
- **D4 movement:** how the words of a sentence are put together.
- **D5 naming:** Augustine's picture, the degenerate case.

§§33–64 add five more:

- **D6 · Bearer.** A name's bearer is one of its relations: the one under the naming kind. When the bearer is gone, that relation loses its object, and

every other relation of the word remains (§§39–45). Meaning is not the bearer. Meaning is what the word relates to, and a destroyed sword relates to much besides its blade.

- **D7 · Sample.** A sample is a token of the language, not a thing the language describes (§50, §53). Examples are the pointing that goes with `this` and `there`, and the colour chip shown with *sepia*. It holds a post like a word does. Nothing is measured *against* it inside the game, because it is what the measuring is done *with*.
- **D8 · Place.** A word's condition may be set by the words before it: in the squares game, the square the reader stands on (§48). Whether a word is simple or composite is a relation the word holds (its *width*), read off use. It is not decided in advance (§§46–47, §64).
- **D9 · Rule from practice.** A rule is what the counts say most often across play. A mistake is a minority count, voted down without being marked. A mistake that is seen corrected is played in a game of its own, so it becomes a condition rather than a deletion (§54).
- **D10 · Analysis.** A sentence about a composite ("the broom") and its analysis into parts ("the stick, and the brush fitted on it") are two games. Neither is the hidden meaning of the other (§§60–64). In the squares game, `U` is not an abbreviation of `R G`. It is one word, with posts at two offsets and a width of two.

2. The remarks, in Phient's terms

§§33–36 — Attending to the shape, or to the colour

The remarks. Someone might say you must already know the game before a pointing can teach you, but also that you must *attend* to the shape or to the colour when you point. What is attending to the colour? We do various things. No single inner act is always there (§33). Even someone who attends to the shape in a particular way may take the explanation differently (§34). There are characteristic experiences of pointing at a shape, but they do not occur every time. Even if they did, what makes it pointing at the shape is the circumstances: what went before and what follows (§35). Where no bodily act can be named, we say a mental act corresponds to the word. The grammar leads us there (§36).

In Phient's terms. In Chapter I's sepia test, "attending to the colour" was nothing inside the learner. It was the word `colour`, said beforehand, stationing the new word at the colour post. The same pointing without it told the learner every aspect at once. The circumstance that made it a pointing at colour was one more token in the episode.

§§37–38 — The relation between name and thing named

The remarks. What is the relation between a name and what it names? In the builders' game, it consists in things like this: hearing the name brings the thing to mind, the name is written on the thing, or the name is said while pointing at it (§37). Naming looks like a strange, almost occult connection. `this` is not a name, although it is used in pointing out. Philosophical problems arise when language goes on holiday (§38).

In Phient's terms. The relation *is* the count and the function, both read off use. Chapter I's toolbox found that `this` holds no post at all: the pointing gesture beside it does. So the counts agree with §38 that `this` is no name.

§§39–45 — Excalibur, and Mr N. N.

The remarks. Why would one want to make `this` a name? Because a name should stand for something simple. `Excalibur` names a sword made of parts. Once the sword is broken, "Excalibur has a sharp blade" still makes sense (§39). The word *meaning* is misused when it stands for the thing that corresponds to the word. When Mr N. N. dies, the bearer dies, not the meaning (§40). A sign whose tool is broken still has a place in the game: B shakes his head (§41). For a large class of cases, the meaning of a word is its use in the language. The meaning of a name is sometimes explained by pointing to its bearer (§43). `this` can never be without a bearer, but that does not make it a name (§45).

In Phient's terms (D6). Phical stores no point, so a word's table row never held the bearer. It holds the word's relations. Remove the bearer and one relation, the one under the naming kind, now points at nothing. Every other relation stays: what the name is used with, what it is said under, what it moves. That is §40 in the representation: the row survives the thing. This chapter does not measure it; a quest that asks Phi-4 about a named thing before and after it is gone would.

§§46–49 — Simples

The remarks. Theaetetus, and Wittgenstein's own *Tractatus*: the primary elements can only be named (§46). But what are the simple parts of a chair: the pieces of wood, or the molecules? "Simple" and "composite" have no meaning outside a particular game (§47). The game of §48: a sentence describes an arrangement of coloured squares, read row by row, one letter per square, R G W B. Is a square simple or composite? It depends on what we are doing. A name can itself be a sentence in another game (§49).

In Phient's terms (D8). We play §48 on a 3 × 3 grid with §64's addition: four more words, each naming two squares side by side (U = R G, V = W B, X = G W, Y = B R). The learner is told only that a word covers squares next to each other in a row, one to three of them. It is not told which words are composite or how wide any word is. Each word's condition is where the reader stands, and that is set by the words before it. §2 below reports the measurements.

§50 — The standard metre

The remarks. One cannot say of the standard metre in Paris that it is, or is not, one metre long. It is not something described in the game. It is an instrument of the language, a means of representation (§50).

In Phient's terms (D7). Our pointing tokens (`⟨point:p3⟩`) and the sepia chip are such instruments. In the toolbox (Chapter I) they hold posts like words do: `⟨point:p3⟩` moves *place* to p3. Nothing in the game measures them against a place. They are what the place is given by.

§§51–56 — Correspondence, and rules read off practice

The remarks. What does the correspondence between a sign and a colour consist in? It is shown by the practice of the language (§51, §53). There are many ways a rule can figure in a game. It can be taught, or looked up in a table. Or it can be used nowhere, and an observer reads it off the practice, like a law of nature. How does the observer tell a mistake from correct play? By characteristic signs, such as the player correcting a slip (§54). §§55–56 ask what a name must stand for if it is to survive its bearer. A remembered sample is still a sample.

In Phient's terms (D9). Every learner in these games is §54's observer. No learner is given a rule; it reads rules off the practice. §3 measures two cases: a practice with mistakes nobody marks, and a practice with mistakes that are seen corrected.

§§57–59 — Red things destroyed, and red

The remarks. Something red can be destroyed, but red cannot, so the meaning of *red* seems independent of any red thing (§57). A sentence like "red exists" is really about the use of a word (§58). We see parts, not the "elements of reality" that names were supposed to stand for (§59).

In Phient's terms. *red* is a word with posts. It is not a stored colour. Its relations are counts over episodes, and no episode stores a thing, so destroying every red thing in a new episode changes nothing already counted.

§§60–64 — The broom, and the composite squares

The remarks. "My broom is in the corner" can be taken apart into "the broomstick is there, and the brush is there, and the stick is fixed in the brush". But the second is not what the first *really* meant (§60). Orders for composite objects and orders for their parts make two games (§§60–62). The analysed form is not more fundamental; it loses something as well as showing something (§63). §64 changes the squares game of §48 so that there are names for rectangles made of two squares. Is that the same game? It is a different game, and the new names are not abbreviations.

In Phient's terms (D10). In the squares game, `U` is not stored as "R followed by G". It is learned as one word with a width of two, relating to R at its first square and G at its second. The learner never sees `U` split into parts. It reads `U` where it stands.

3. The squares game (§48, §64): measured

`bin/squares_game`. Arrangements are random, and each sentence describes one. Composite names are used, where they fit, half the time in training. Training sentences contain at most one composite, and never in the middle column. There are four tests of 300 sentences each:

1. plain names only;
2. one composite at a place seen in training;
3. one composite at a **new** place (the middle column);
4. **two or more** composites in one sentence (never seen in training).

The learners, all trained on the same sentences:

- **lookup** (whole sentence remembered);
- **naming table** (each word names the colour it most often goes with at its own index in the sentence; the sentence is the list, §1);
- **positional softmax** (one classifier per square over `word@index` features: the standard position-indexed model);
- **Phient reader** (D8). Each word relates under two kinds, *width* and *colour at offset o*. They are counted as expected counts over every way the sentence can lie on the grid, each way weighted by how well the current relations fit it, and counted again until they settle (EM, 12 rounds).

Exact match (all nine squares), seed 7 / seed 8:

learner	training sentences	plain	composite, trained place	composite, new place	two or more composites
naming table	50	100 / 100%	0.7 / 1.3%	3.0 / 5.7%	0 / 0%
positional softmax	50	15.3 / 16.3%	1.3 / 3.0%	0.7 / 0.7%	0 / 0%
Phient reader	**50**	**100 / 100%**	**100 / 100%**	**100 / 100%**	**100 / 100%**
positional softmax	1,000	50.0 / 64.3%	17.0 / 16.7%	2.7 / 5.0%	0 / 0%
Phient reader	1,000	100 / 100%	100 / 100%	100 / 100%	100 / 100%

Lookup scores 0–0.3% everywhere, and the naming table does not change with more data.

What each word came to relate to (reader, 1,000 sentences, both seeds):

word	P(width 1 / 2 / 3)	colour at offset 0	colour at offset 1
`R`, `G`, `W`, `B`	1.00 / 0.00 / 0.00	itself (1.00)	—
`U`	0.00 / 0.99 / 0.00	R (1.00)	G (1.00)
`V`	0.00 / 0.99 / 0.00	W (1.00)	B (1.00)
`X`	0.00 / 0.99 / 0.00	G (1.00)	W (1.00)
`Y`	0.00 / 0.99 / 0.00	B (1.00)	R (1.00)

What it shows, and what it does not.

- **Simple and composite are read off use (§47, §64).** The reader was allowed widths of 1, 2 or 3. It found width 1 for the plain names and width 2 for the composites, with the right colour at each offset, from 50 sentences.
- **A condition set by the words before it generalises where position-indexed features cannot.** The positional softmax ties each word to the index it happened to have. One composite earlier in the sentence shifts every later word, and the classifier does not see the shift as a shift. It fails on new places and on combinations, and even plain sentences fall to 50–64% at 1,000 sentences. The reader never indexes: it stands somewhere, and each word moves it on.
- **This is not a fair fight with a modern sequence model,** and we say so. The reader is told the form of the game: words in reading order, each covering squares next to each other in a row. A learner with learned alignment (attention, CTC) would likely find the same. The honest comparison is with a position-indexed model, the naming picture and lookup. The claim is about *conditions*: once "where the reader stands" is the condition, D1–D4 need nothing more.

4. A teacher who makes mistakes (§54): measured

`bin/language_game mistakes`. The builders' game of Chapter I, with 3,600 training episodes. In a share p of them, one aspect of what followed is replaced by another value that aspect takes. In **silent** mistakes, nothing marks the error. In **corrected** mistakes, the error is seen corrected, so the episode is played in a game of its own, `corrected` (D9). Tests are the four rule tests of Chapter I.

Trial 8 (the first version). Silent mistakes did no harm. With up to 40% of episodes wrong in one aspect, Phical counts stayed at 100% in every column. phinet and two-speed stayed between 97.9% and 100%, and bag of words between 99.8% and 100%. The majority carries the rule.

Corrected mistakes were *worse*. Bag of words fell to 89–95% at 30–40%, and on seed 8 at 10% the counts fell to 77.8%. The fall equalled the report test's share. The cause was D3. Whether a word may be carried from one game into another was judged across *all* games together, and the `corrected` game, full of errors by design, broke that agreement for one aspect. So words were no longer carried from orders into reports.

Trial 9: agreement is judged between the two games concerned. A word told in game *g* may be read in game *h* under an aspect if *g* and *h* agree on it. A third game has no say. Where two games share no firm word, the language's own agreement decides.

Exact match on all four rule tests, seed 7 / seed 8, after trial 9:

Silent mistakes (share of episodes wrong in one aspect):

learner	0%	10%	20%	30%	40%
lookup	47.2 / 47.0	46.7 / 45.8	45.9 / 44.6	43.8 / 43.6	43.2 / 41.4
naming table	64 / 64	64 / 64	64 / 64	64 / 64	64 / 64
bag of words	100 / 100	100 / 99.8	100 / 99.8	99.9 / 100	100 / 99.9
Phical counts, movement	**100 / 100**	**100 / 100**	**100 / 100**	**100 / 100**	**100 / 100**
Phient (phinet, R = 32)	100 / 100	97.9 / 100	100 / 100	100 / 100	100 / 100
two speeds, N = 32	100 / 100	97.9 / 100	100 / 100	100 / 100	100 / 100

Corrected mistakes (played in a game of their own):

learner	10%	20%	30%	40%
lookup	47.0 / 46.2	46.3 / 46.1	45.4 / 45.7	45.3 / 44.4
naming table	64 / 64	64 / 64	64 / 64	64 / 64
bag of words	99.6 / 100	97.0 / 97.5	89.8 / 94.6	88.7 / 94.5
Phical counts, movement	**100 / 100**	**100 / 100**	**100 / 100**	**100 / 100**
Phient (phinet, R = 32)	100 / 98.6	100 / 100	95.9 / 100	98.2 / 100
two speeds, N = 32	100 / 98.6	100 / 100	95.9 / 100	100 / 100

- **Counts read the rule off the practice** in every column on both seeds: silent mistakes up to 40%, and corrected ones.
- **The functions follow within 1–4 points.** Where they slip, the counts do not.
- **Bag of words is hurt by corrections, not by silent mistakes.** Its corrected episodes still train the same word weights, and the `corrected` feature cannot take the whole blame. Trial 9 does not touch it, because bag of words has no D3.

5. What Chapter II adds

- Two more games in which Phient's paradigm, taken literally, does what the remarks say language does:
  - a word's condition set by the words before it (§48);
  - simple and composite read off use (§47, §64);
  - a rule read off practice that contains mistakes (§54).

- One more fault found and fixed (trial 9). It is the same kind of fault as Chapter I's: a judgement made over the whole language that belonged to the particular games concerned. Both chapters point the same way: in Phient, the condition is local.
- For the Phi-4 student, §54 matters most. Phi-4 is a teacher that makes mistakes. Silent errors in up to 40% of episodes, in one aspect each, do not move a counting learner. When the errors are marked, they must be kept as a condition of their own, and judged only against the games they touch.

6. Next

- **Run Phi-4 in the builders' game** (`scripts/phi4_builders.ps1`). §54 predicts how much its errors will cost a student, and the teacher report will show which aspects it gets wrong.
- **Descriptive Logic III, §§65–88:** family resemblance, games without a common essence, blurred edges, and rules that do not cover every case. The posts table gives a way to measure family resemblance: words that share posts under overlapping, not identical, sets of kinds.
- **Bearers (D6), measured:** a quest that asks Phi-4 about a named thing before and after it is gone.

Code:

- `phiano/src/bin/squares_game/{world, reader, baselines, main}.rs`;
- `phiano/src/bin/language_game/{experiments, compose, learners}.rs`.

Raw tables: docs/104_*.md (squares game, mistakes trial 8 and trial 9, and a determinism rerun).

Run:

- `cargo run --release --bin squares_game -- out.md --seed 7` (under a second);
- `cargo run --release --bin language_game -- mistakes out.md --seed 7` (about 35 minutes).